i percent be the interest on one rupee per year and P be the present value of the annuity. In the case of immediate annuity, payments are made periodically at the end of specified period. Since the first installment is paid at the end of first year, its present value is $\frac{a}{1+i}$, the present value of second installment is $\frac{a}{(1+i)^2}$ and so on. If the present value of last installment is $\frac{a}{(1+i)^n}$, then we have

$$P = \frac{a}{1+i} + \frac{a}{(1+i)^2} + \frac{a}{(1+i)^3} + \cdots \frac{a}{(1+i)^n}$$

$$= \frac{a}{(1+i)^n} + \frac{a}{(1+i)^{n-1}} + \ldots + \frac{a}{(1+i)}$$

$$= \frac{a}{r^n}\left[1 + r + r^2 + \ldots + r^{n-1}\right], \text{ taking } 1 + i = r$$

$$= \frac{a}{r^n}\left[\frac{r^n - 1}{r - 1}\right], \text{ G.P with common ratio, } r > 1$$

$$= \frac{a}{(1+i)^n}\left[\frac{(1+i)^n - 1}{1 + i - 1}\right]$$

$$= \frac{a}{i}\left[1 - \frac{1}{(1+i)^n}\right]$$

$$\boldsymbol{P = \frac{a}{i}\left[1 - \frac{1}{(1+i)^n}\right]}$$

(iii) Amount of annuity due at the end of *n* period

Annuity due is an annuity in which the payments are made at the beginning of each payment period. The first installment will earn interest for n periods at the rate of 'i' percent per period. Similarly second installment will earn interest for $(n - 1)$ periods, and so on the last interest for on period.

$$\therefore A = a(1+i)^n + a(1+i)^{n-1} + \ldots + a(1+i)^1$$

$$= a(1+i)[(1+i)^{n-1} + (1+i)^{n-2}) + \ldots + 1]$$

$$= a(1+i)[1 + (1+i) + (1+i)^2 + \ldots + (1+i)^{n-1}]$$

$$= ar[1 + r + r^2 + \ldots + r^{n-1}] \text{ , put } 1 + i = r$$

$$= ar\left[\frac{r^n - 1}{r - 1}\right], \text{ G.P with common ratio, } r > 1$$

$$= a(1+i)\left[\frac{(1+i)^n - 1}{1 + i - 1}\right]$$

$$= \frac{a(1+i)}{i}[(1+i)^n - 1]$$

$$\boldsymbol{A = \frac{a(1+i)}{i}[(1+i)^n - 1]}$$

(iv) Present value of annuity due

Since the first installment is paid at the beginning of the first period (year), its present value will be the same as 'a', the annual payment of annuity due. The second installment is paid in the beginning of the second year, hence its present value is given by $\frac{a}{(1+i)}$ and so on. The last installment is paid in the beginning of nth year, hence its present value is given as $\frac{a}{(1+i)^{n-1}}$

If P denotes the present value of annuity due, then

$$P = a + \frac{a}{1+i} + \frac{a}{(1+i)^2} + \frac{a}{(1+i)^3} + \ldots + \frac{a}{(1+i)^{n-1}}$$

$$= a\left[1 + \frac{1}{1+i} + \frac{1}{(1+i)^2} + \frac{1}{(1+i)^3} + \ldots + \frac{1}{(1+i)^{n-1}}\right]$$

$$= a\left[\frac{1 - \left(\frac{1}{1+i}\right)^n}{1 - \frac{1}{1+i}}\right]$$

$$= a\left[\frac{\frac{(1+i)^n - 1}{(1+i)^n}}{\frac{(1+i) - 1}{1+i}}\right]$$

$$= \frac{a(1+i)}{i}\left[\frac{(1+i)^n - 1}{(1+i)^n}\right]$$

$$\boldsymbol{P = \frac{a(1+i)}{i}\left[1 - \frac{1}{(1+i)^n}\right]}$$

(v) Perpetual Annuity

Perpetual annuity is an annuity whose payment continuous for ever. As such the amount of perpetuity is undefined as the amount increases without any limit as time passes on. We know that the present value P of immediate annuity is given by

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